🔥 FIREWISE-X — 3-Minute Video Presentation Script

NASA International Space Apps Challenge 2026 | Challenge: Flame in Freefall

Target Duration: 180 Seconds (3:00 Min) **Spoken Word Count:** 408 Words (~136 wpm) **Video Resolution:** 1080p 60fps

Platform URL: http://localhost:3000 **Data Source:** NASA PSI-25 (BASS-II)

🎬 **Director & Presenter Delivery Guidelines:** Keep delivery energetic, scientific, and confident. Use a smooth mouse cursor with slight yellow highlight circle during screen recording. Pacing is calibrated at ~135 words per minute. Practice speaking along with the visual clicks so each scene transitions exactly at the indicated timestamp.

TIMECODE	VISUAL SCREEN ACTION (WHAT TO SHOW)	VOICEOVER DIALOGUE (WHAT TO SAY)	ON-SCREEN GRAPHICS & SFX
0:00 – 0:25 25 SEC	**Home Page & Science Primer** • Open on `http://localhost:3000` Home. • Zoom into Hero text: "*Understand the Flame. Follow the Evidence.*" • Scroll smoothly down to the **Earth (1g) vs ISS (0g)** visual comparison card. • Hover mouse over the buoyant flame vs spherical space dome.	"In space, fire doesn't behave like it does on Earth. Without gravity, natural buoyancy vanishes—hot gases cannot rise away. Instead, flames morph into quiet, creeping spheres. But this calm appearance hides a lethal reality: sluggish cabin airflow causes severe incomplete combustion, generating toxic Carbon Monoxide surges up to **950 ppm**! Welcome to **FIREWISE-X**: an evidence-grounded research engine transforming 129 real NASA ISS combustion tests into life-saving spacecraft fire safety protocols." *Tone: Dramatic, authoritative hook; steady pace.*	**Text Overlay:** "FIREWISE-X" "NASA BASS-II (PSI-25)" **SFX:** Subtle low synth drone; atmospheric spacecraft hum.
0:25 – 0:55 30 SEC	**Mission Control Live Telemetry** • Click 🚀 **Mission Control** in top nav. • Click pill: 🔴 **TEST_O13A** (PMMA Rod). • Point cursor to the **Cabin Toxicity Threat Gauge** swinging into the red lethal zone. • Hover over the canvas chart: show glowing crosshair and floating HUD box with time, O_2, and 950 ppm CO.	"Stepping into Mission Control, we interface directly with real flight data from NASA's BASS-II experiment on the International Space Station. Selecting Test O13a—a half-inch acrylic rod—our dynamic Cabin Threat Meter immediately flags a critical hazard. As we trace the high-DPI telemetry stream, our interactive HUD reveals the exact second oxygen plummets and Carbon Monoxide spikes to an alarming **950 ppm**—far exceeding safe astronaut thresholds." *Tone: Focused, urgent, pointing out telemetry spikes.*	**Text Overlay:** "Peak Toxic CO: 950 ppm" "Threshold: >500 ppm Lethal" **SFX:** Interface click blip; soft warning chime.
0:55 – 1:30 35 SEC	**129 Tests Explorer & Catalog** • Click 📊 **129 Tests Explorer**. • Click preset: 🚨 **Critical CO Spikes (>500 ppm)**	"Unlike projects built on synthetic mocks, FIREWISE-X directly ingests all 129 flight combustion tests from NASA's Physical Sciences Informatics archive. With one click on our preset filters, researchers instantly isolate all runs exceeding 500 ppm CO. We can seamlessly toggle from dense tabular data to mobile-friendly visual cards, and open an	**Text Overlay:** "129 Real Spaceflight Runs" "NASA DOI: 10.60555/4qc4-de67" **SFX:** Crisp filter click; modal pop whoosh.

TIMECODE	VISUAL SCREEN ACTION (WHAT TO SHOW)	VOICEOVER DIALOGUE (WHAT TO SAY)	ON-SCREEN GRAPHICS & SFX
	. • Click button: 📱 **Switch to Visual Cards** . • Click on **TEST_M5** card to pop open the **Test Detail Modal** with 17 telemetry fields. • Close modal (click ✕).	in-depth parameter modal detailing fuel thickness, airflow duct restrictions, and investigator observations." *Tone: Energetic; showing depth and real data provenance.*	

TIMECODE	VISUAL SCREEN ACTION (WHAT TO SHOW)	VOICEOVER DIALOGUE (WHAT TO SAY)	ON-SCREEN GRAPHICS & SFX
1:30 – 2:05 35 SEC	**Flame Studio & Computer Vision** • Click 🎥 **Flame Studio (CV)** . • Play native ISS video (Test B1). • Point to the right-hand canvas: highlight golden segmented flame, cyan bounding box, red centroid. • Move Luma Threshold slider (140 → 160). • Click 📈 **Scan 20 Video Frames** button. • Watch the temporal area curve render.	"In Flame Studio, we analyze native H.264 video captured inside the ISS Microgravity Science Glovebox. Using classical Rec.709 luminance thresholding, our algorithm tracks flame front boundaries, bounding dimensions, and centroid trajectories in real-time. By clicking 'Scan Video Frames', FIREWISE-X automatically samples temporal keyframes to graph the complete flame front expansion curve. And maintaining strict scientific honesty, we abstain from uncalibrated 3D millimeter claims, reporting strictly in verifiable camera pixels." *Tone: Technical precision, emphasizing scientific integrity.*	**Text Overlay:** "Classical Rec.709 Luma CV" "Temporal Flame Front Growth" "Honest Pixel Disclosure" **SFX:** Camera shutter click; smooth curve draw sweep.
2:05 – 2:35 30 SEC	**Sensors Anomaly & Evidence Engine** • Click 📈 **Sensors & Anomaly** . • Adjust Z-Threshold slider (3.0σ). • Show glowing red anomaly beacon rings on the surge points. • Click 📚 **Evidence Engine** in nav. • Filter by 'Toxic Emissions'. • Hover over NASA TM-20210011385 card and click DOI link.	"To detect anomalies without deep-learning hallucination, our Sensors tab implements a Robust Rolling Z-score powered by Median Absolute Deviation (MAD), pinpointing sudden toxic surges while rejecting sensor noise. Every detection is fully transparent. In our Evidence Engine, every algorithm and telemetry spike maps directly to peer-reviewed NASA literature and Technical Memorandums with exact DOIs and explicit limitation boundaries." *Tone: Analytical, establishing peer-reviewed credibility.*	**Text Overlay:** "Robust Z-Score: Median / MAD" "Zero AI Hallucination" "Direct NTRS DOI Citations" **SFX:** Anomaly ping sound; page flip sound.
2:35 – 3:00 25 SEC	**Compare Runs & Mission Impact** • Click ⚖️ **Compare Runs** in nav. • Select Run A (Test B1 Thin Film) vs Run B (Test M5 Thick Slab). • Highlight oxygen depletion (-3.2%) and CO spike (500 ppm). • Scroll to footer and show final NASA Space Apps 2026 title graphic.	"Finally, Compare Runs reveals critical physical contrasts: thick acrylic slabs sustain two-sided burning that consumes over 3% oxygen and pumps out 500 ppm CO, whereas thin films extinguish rapidly when airflow ceases. By turning archived flight telemetry into transparent, explainable intelligence, FIREWISE-X provides the foundation for autonomous fire safety aboard Artemis, Gateway, and deep-space habitats. **Understand the flame. Follow the evidence.** Thank you." *Tone: Inspiring, impactful conclusion with strong sign-off.*	**Text Overlay:** "Artemis & Gateway Ready" "Understand the Flame." "Follow the Evidence." **SFX:** Uplifting cinematic music swell; final fade out.